

Quiz 4

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (3 points) Find the velocity vector $\mathbf{v}(t)$ and the position vectors $\mathbf{r}(t)$ of a particle that has $\mathbf{a}(t) = \langle t, t^2, 1 \rangle$, $\mathbf{v}(0) = \langle 0, 0, 1 \rangle$ and $\mathbf{r}(0) = \langle 1, 0, 0 \rangle$.

$$\begin{aligned}\mathbf{v}(t) &= \mathbf{v}(0) + \int_0^t \mathbf{a}(\tau) d\tau = \langle 0, 0, 1 \rangle + \int_0^t \langle \tau, \tau^2, 1 \rangle d\tau \\ &= \langle 0, 0, 1 \rangle + \langle \frac{\tau^2}{2}, \frac{\tau^3}{3}, \tau \rangle = \langle \frac{t^2}{2}, \frac{t^3}{3}, t+1 \rangle\end{aligned}$$

$$\begin{aligned}\mathbf{r}(t) &= \mathbf{r}(0) + \int_0^t \mathbf{v}(\tau) d\tau = \langle 1, 0, 0 \rangle + \int_0^t \langle \frac{\tau^2}{2}, \frac{\tau^3}{3}, \tau+1 \rangle d\tau \\ &= \langle 1, 0, 0 \rangle + \langle \frac{\tau^3}{6}, \frac{\tau^4}{12}, \frac{\tau^2}{2} + \tau \rangle \\ &= \langle \frac{t^3}{6} + 1, \frac{t^4}{12}, \frac{t^2}{2} + t \rangle\end{aligned}$$

Problem 2. (4 points) Let $\mathbf{r}(t) = e^t \mathbf{i} + \sqrt{2}t \mathbf{j} + e^{-t} \mathbf{k}$ that defines a space curve.

(a) Find the unit tangent vector $\mathbf{T}(t)$ of the curve at the point with $t = 0$.

$$\begin{aligned}\mathbf{r}'(t) &= \langle e^t, \sqrt{2}, -e^{-t} \rangle \\ \Rightarrow \mathbf{r}'(0) &= \langle e^0, \sqrt{2}, -e^0 \rangle = \langle 1, \sqrt{2}, -1 \rangle, \quad |\mathbf{r}'(0)| = \sqrt{1^2 + (\sqrt{2})^2 + (-1)^2} = 2 \\ \Rightarrow \mathbf{T}(0) &= \frac{\mathbf{r}'(0)}{|\mathbf{r}'(0)|} = \frac{1}{2} \langle 1, \sqrt{2}, -1 \rangle = \langle \frac{1}{2}, \frac{\sqrt{2}}{2}, -\frac{1}{2} \rangle\end{aligned}$$

(b) Find the length of the arc defined by $\mathbf{r}(t)$ with $0 \leq t \leq 1$.

$$\begin{aligned}|\mathbf{r}'(t)| &= \sqrt{(e^t)^2 + (\sqrt{2})^2 + (-e^{-t})^2} = \sqrt{e^{2t} + 2 + e^{-2t}} = \sqrt{(e^t + e^{-t})^2} = e^t + e^{-t} \\ \Rightarrow L &= \int_0^1 |\mathbf{r}'(t)| dt = \int_0^1 e^t + e^{-t} dt \\ &= [e^t - e^{-t}]_{t=0}^{t=1} = (e - e^{-1}) - (1 - 1) = e - \frac{1}{e}\end{aligned}$$

Problem 3. (3 points) Let $\mathbf{r}(t) = \langle 5t, 2\sin t, 2\cos t \rangle$. Find the curvature of the curve.

$$\begin{aligned}\mathbf{r}'(t) &= \langle 5, 2\cos t, -2\sin t \rangle \Rightarrow |\mathbf{r}'(t)| = \sqrt{5^2 + (2\cos t)^2 + (-2\sin t)^2} = \sqrt{29} \\ \mathbf{T}(t) &= \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \frac{1}{\sqrt{29}} \langle 5, 2\cos t, -2\sin t \rangle \\ \mathbf{T}'(t) &= \frac{1}{\sqrt{29}} \langle 0, -2\sin t, -2\cos t \rangle \Rightarrow |\mathbf{T}'(t)| = \frac{1}{\sqrt{29}} \sqrt{0^2 + (-2\sin t)^2 + (-2\cos t)^2} \\ &= \frac{2}{\sqrt{29}} \\ \Rightarrow \kappa &= \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|} = \frac{\frac{2}{\sqrt{29}}}{\sqrt{29}} = \frac{2}{29}\end{aligned}$$